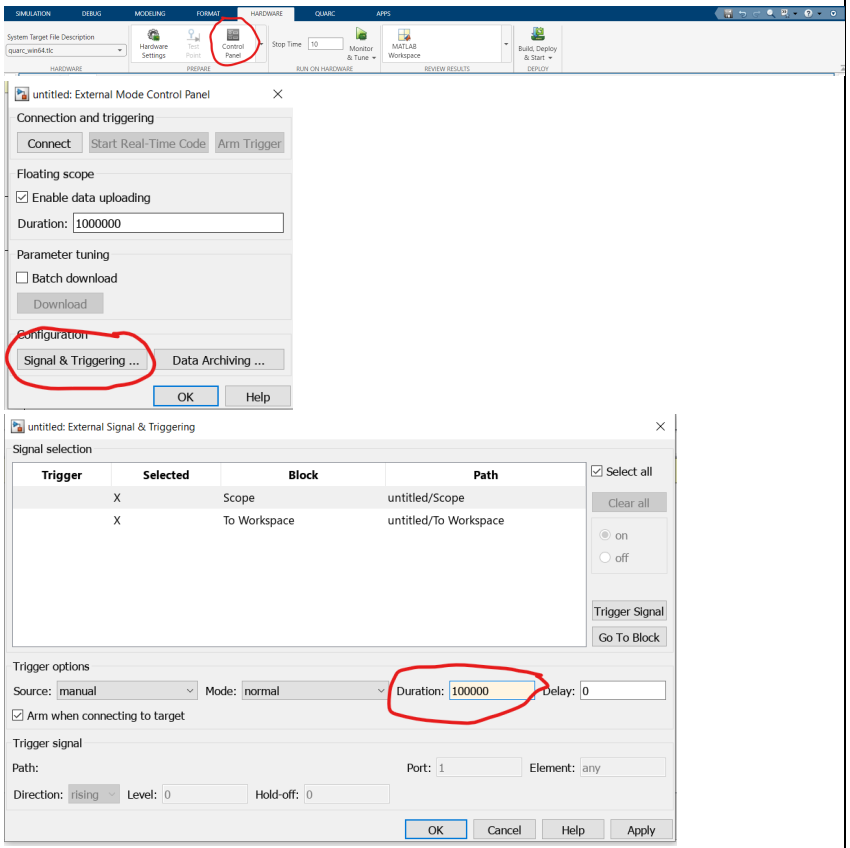
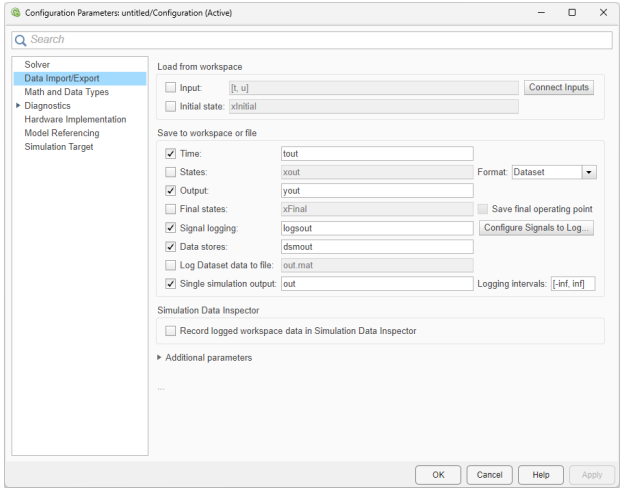
Common Issues and Fixes for 561/661 and 562/662 Labs

Last updated: September 2026. If you hit a problem that is not listed here, tell your TA so it can be added.

Symptom	Likely cause	Fix	Applies to
To Workspace returns only one value	The To Workspace block stores a fixed number of points (default 1000). When the run length puts the last sample at point 1001, the buffer clears and only the final value is kept.	Either shorten the run by one time step (10 s → 9.99 s), or raise the stored-point limit: Hardware tab → Control Panel → Signal & Triggering → increase Duration. 	All
Error "The card specific option specified is not recognized."	The Board-specific options field in HIL Initialize contains an entry that is not valid for this board.	Clear the field completely and run the model again.	All
MATLAB licence validation failed	Your laptop was not fully connected to Wi-Fi when MATLAB tried to confirm the licence. Usually shown together with a server that cannot be reached.	Connect to Wi-Fi, then open MATLAB again.	All
Model outputs disagree with the hardware	Solver left on Auto. A variable-step solver takes steps that do not align with the fixed	Model Settings → Solver → Type: Fixed-step, Solver: ode1 (Euler), step size equal to the HIL sample time.	All

Symptom	Likely cause	Fix	Applies to
	sample time of the QUARC blocks, so the ODE, transfer-function and filtered signals drift from the HIL Read output. The filtered speed often appears scaled down.	Report the solver and step size you used. If your Auto and fixed-step results differ, include both and discuss the difference where you compare your models against the hardware. Your lab manual carries the same note.	
HIL Read does not output the expected signal	Wrong channel number or channel type. The output usually looks like random noise around zero.	Check the channel number and the channel type against the sensor being read.	All
Error "Unable to set rate base rate. A value is outside the valid range"	The requested step size is smaller than the DAQ can handle.	Set the step size to 0.001.	All
Tachometer gives a constant signal that does not vary with speed	Usually a bad cable between the tach port and the UPM.	Tell your TA — the cable is the usual cause and they will replace it.	562/662
Warning "Out not created"	With Single simulation output enabled, the To Workspace block writes to out.simout, which can fail with the Quanser hardware.	Uncheck Single simulation output. The block then writes to simout directly. 	All
The model halts while you are deliberately stalling the motor	The stall-detection block stops the model when the shaft is stationary for more than 10 s with more than 5 V applied. Some labs ask you to hold the shaft still on purpose, so the block trips if you hold too long or exceed 5 V.	Keep applied voltages at or below 5 V, hold for only a couple of seconds per reading, and release between readings. The block is protecting the motor from overheating — it is not a fault, and you must not remove it. Work within its limits and take each reading quickly.	561/661
The speed signal is full of spikes	Not a fault. This is a real property of the measurement chain, and your lab manual asks you to explain why it happens.	**There is nothing to fix here.** To work out the explanation, put the encoder position on its own Scope and zoom in on it before it reaches the derivative block — this is the hint given in the manual. The low-pass filter you add later is what reduces the spikes.	561/661
QUARC build fails, or the	quarc_win64.tlc not selected, or missing	QUARC tab → QUARC Targets → quarc_win64. If it is missing, QUARC is not	All

Symptom	Likely cause	Fix	Applies to
target is not available	from the QUARC Targets list. Also occurs after a model is moved or renamed.	installed for that MATLAB version — move to another laptop and tell your TA the machine number. Save your model in a local folder whose path has no spaces before building.	
Encoder angle is wrong by a constant factor	Wrong counts-to-angle conversion factor.	2048 counts per revolution: degrees = counts × 360/2048; radians = counts × 2π/2048. A factor-of-four error means you used 512 lines per revolution instead of 2048 quadrature counts.	All